AI-Based Cognitive Gaming & Memory Assistance Platform for Elderly Dementia Patients in the North Eastern Region (NER)

Cognitive gaming basically means games designed to exercise specific thinking abilities of the brain.

Mental processes like:

- memory
- attention
- reasoning
- language
- problem-solving
- processing speed
- spatial understanding

The game is designed so that, while playing, the person repeatedly uses one or more of these abilities.

- A matching-pairs game exercises memory.
- A game where the user quickly taps a target object exercises attention and reaction speed.
- A game where they arrange steps in the correct order exercises planning/executive function.
- Naming an object from a picture exercises language and word recall.
- Finding a route through a simple maze exercises visuospatial ability.

For an elderly dementia/MCI-focused app, the idea is- give the person short, manageable mental activities that keep them cognitively engaged and also let the system track how their performance changes over time.

So if someone repeatedly plays a memory game, you might record:
Accuracy: 80%
Time taken: 35 seconds
Wrong attempts: 3
Hints used: 1

Then next week:

Accuracy: 75%
Time taken: 42 seconds
Wrong attempts: 5
Hints used: 2

Those games therefore have two roles in your project:

1. Training/engagement: the elderly user actively exercises memory, attention, etc.
2. Measurement: the app gets numerical data such as accuracy, reaction time and errors that can later be used for personalization and trend monitoring.

The landmark LASI-DAD nationwide study (published in *Alzheimer's & Dementia*, 2023) estimates dementia prevalence for adults ages 60+ in India at 7.4%, meaning about 8.8 million Indians older than 60 live with dementia. Critically for a project pitch, this burden is projected to grow sharply: researchers estimate an increase from 8.8 million individuals aged 60 and older with dementia in 2016 to 16.9 million in 2036 — nearly doubling in two decades.

The same study found considerable variation across the country, with lowest prevalence in Delhi at 4.5% and highest in Jammu and Kashmir at 11.0%. Two risk gradients drive this and both apply heavily to rural NER: significant age and education gradients, sex and urban/rural differences. Specifically, prevalence is higher among women (nearly double men's), among rural populations, and among those with low/no formal education

LASI-DAD studied people across 18 states, but from the entire North-East, **only Assam was included**. So for states like Nagaland, Manipur, Meghalaya, Mizoram, Tripura, Arunachal Pradesh, and Sikkim, we have much less reliable data about:

- how common dementia is,
- which groups are most affected,
- what local risk factors matter,
- and how dementia may present across different languages/cultures.

Fast-ageing, out-migration-heavy demographics — younger people migrate for work, leaving elderly relatives with thinner caregiving support.
Extreme linguistic diversity — 200+ languages/dialects (Assamese, Bodo, Khasi, Mizo, Manipuri/Meitei, Nagamese, Nyishi, Kokborok, etc.). Almost every existing app is English/Hindi-only.
Low awareness and high stigma — nationally, low awareness and stigma associated with dementia is highly prevalent among both the general public and health professionals. Dementia is often dismissed as "normal old age." Stride-dementia
Sparse specialist care — few neurologists/geriatricians, patchy connectivity in hill and border districts.
Culturally distinct memory anchors — reminiscence therapy relies on personally/culturally familiar images, music, festivals and food; generic Western content simply doesn't work here.

Prevention/early stage — promising. The long-running ACTIVE trial found that cognitive speed training can reduce the risk of dementia, and results suggest cognitive games could be effective in training cognition if used prior to the onset of dementia. Framing your platform as **early-intervention + risk-reduction** (targeting MCI and mild dementia) is the most defensible position. Dementia Care CentralNature
Mild dementia — feasible and modestly beneficial. A 24-week home RCT using the Constant Therapy app in Alzheimer's patients showed long-term computerized cognitive training is feasible in mild cognitive impairment and mild dementia stages; patients adhered (80% adherence) and improved performance over time, with specific gains in visual/auditory memory, attention and arithmetic.

Design lesson from the literature: don't ship a bare game grid. Existing apps are narrowly focused on cognitive games, which may fail to engage users long term, and often lack diverse content such as arts, physical exercises, literature-based activities, and social interactions — this narrow focus reduces engagement and adherence. A **holistic, multi-domain** design wins.

3. Existing platforms (know your competition)

Global / clinical:

- **Constant Therapy (Brain+/Speech)** — the most clinically-validated; targets dementia, aphasia, stroke, TBI. ~$300/yr. English-centric.
- **Lumosity, CogniFit, BrainHQ (Posit Science)** — consumer brain-training; BrainHQ is the ACTIVE-derived speed-training lineage.
- **MyReha / GameAAL-type tablet+TV programs** — research-grade multi-domain training for nursing homes.

India-specific (your closest reference points):

- **DemClinic** (Nightingales Medical Trust + ARDSI Bangalore) — a telemedicine expert-led cognitive assessment platform for the elderly to increase access to dementia screening, diagnosis, and care. Stride-dementia
- **DemLink App** (NMT) — aims to educate families of loved ones with dementia and provides access to care and support via a mobile app. Stride-dementia
- **Moving Pictures India** (NIMHANS) — digital caregiver-education media, being trialled in English and/or Hindi/Kannada. PubMed Central

4. Feature set to integrate

Group features into four pillars:

A. Cognitive gaming (multi-domain, adaptive)

- Games mapped to cognitive domains: **memory** (match-the-pairs with family photos), **attention/processing speed** (ACTIVE-style useful-field-of-view tasks), **language** (naming, word-finding — the domain with the strongest app evidence), **executive function** (sorting, sequencing), **visuospatial** (path-finding), **arithmetic**.
- **AI-driven adaptive difficulty** — auto-adjust based on performance so tasks stay in the "just-hard-enough" zone (key to adherence).
- Short 15–30 min sessions (matches the ~32 min/day adherence seen in trials).

B. Memory assistance (daily-living support)

- **Face/name recall** aid using family-uploaded photos.
- **Reminders** for medication, meals, appointments (with caregiver-set schedules).
- **Reality orientation** module — date, place, time, weather (evidence-backed as RO therapy).
- **Life-story / reminiscence** — localized NER photo/music/video libraries (festivals like Bihu, Hornbill; regional songs; local landmarks/food) to trigger positive recall.
- **"Where am I / safe-return"** — geofencing + emergency contact for wandering (a major caregiver anxiety).

C. AI layer

- **Adaptive engine** personalizing game difficulty and content.
- **Passive cognitive-decline monitoring** — analyze reaction time, accuracy, error patterns over time to flag deterioration and prompt clinical follow-up (accuracy + latency are the standard trial metrics).
- **Conversational voice assistant** in local languages (huge for low-literacy users — voice removes the reading barrier).
- **Speech/voice analysis** for early linguistic markers.
- **NER-language NLP + TTS/ASR** — the hardest and most valuable technical contribution.

D. Caregiver & clinical portal

- Caregiver dashboard: progress trends, alerts, medication logs.
- Caregiver education content (à la DemLink/Moving Pictures).
- Clinician view + exportable reports; optional teleconsult hook (à la DemClinic).

- Digital screening (adapt **MMSE / MoCA / ACE-III / ICMR Neurocognitive Toolbox**, which are tools developed/adapted to account for the considerable educational and linguistic diversity of the Indian context). [Stride-dementia]

Cross-cutting must-haves for NER: offline-first mode, low-bandwidth sync, large fonts/high-contrast/simple UI, voice-first navigation, and multilingual from day one.

5. Technology stack

Layer	Options
Mobile app	React Native or Flutter (offline-first, cross-platform); tablet-optimized large-touch UI
Games	Unity / Godot for richer games, or lightweight in-app HTML5/Canvas for simple ones
Backend	Node.js / Django / FastAPI; PostgreSQL; sync layer for intermittent connectivity
AI/ML	Adaptive difficulty (reinforcement learning / bandit algorithms); decline-detection (time-series ML, gradient boosting on performance features); on-device inference (TensorFlow Lite / ONNX) for offline & privacy
Speech (local langs)	ASR/TTS — fine-tune multilingual models (Whisper, AI4Bharat's IndicTrans/IndicASR/Indic-TTS which cover Assamese, Manipuri, Bodo, etc.); conversational layer via an LLM
Reminiscence content	Cloud media store (S3-compatible) + CDN; community-sourced localized content pipeline
Location/safety	Geofencing, background location, SMS/push fallback
Data & compliance	End-to-end encryption; India's **DPDP Act 2023** compliance; consent flows for cognitively-impaired users (proxy/caregiver consent)
Infra	Progressive Web App fallback for feature phones/low-end Android; SMS-based reminders where data is absent

6. Real challenges to acknowledge

- **Localization at scale** — building game content, voice, and reminiscence libraries across many NER languages is the core hard problem and cost driver. Start with 2–3 (e.g. Assamese + Manipuri/Meitei + one hill language) and expand.
- **Low digital literacy among elderly** — voice-first and caregiver-mediated use is essential.
- **Connectivity** — offline-first is non-negotiable in hill/border districts.
- **Adherence & the "narrow-games" trap** — build the holistic, socially-engaging design the evidence recommends.
- **Evidence honesty** — market as engagement/QoL/early-intervention, not cure.
- **Validation** — partner with ARDSI Guwahati / a regional medical college (Guwahati Medical College was a LASI-DAD site) for a small feasibility pilot; that turns a hackathon/college project into something publishable.

Component	Needs training data?
Games (memory match, sequencing, naming, arithmetic)	**No** — rule-based logic, content is authored not learned
Reminders, reality orientation, geofencing	**No** — plain app logic
Reminiscence library (photos/music/festivals)	**No** — this is *content curation*, not ML
Adaptive difficulty engine	**Light data** — can start with a simple rule-based algorithm, improve with your own usage telemetry later
Cognitive decline detection (flagging deterioration)	**Yes** — needs labeled longitudinal performance data
Speech/voice assistant in NER languages	**Yes** — needs ASR/TTS training or fine-tuning data
Speech-based dementia screening (research-grade)	**Yes** — needs clinical speech corpora

For NER-language speech/NLP:

- **AI4Bharat** (IIT Madras) publishes open ASR/TTS/translation datasets and models covering many Indic languages including Assamese, Bodo, Manipuri to varying degrees — check current coverage since it expands often (this is worth a fresh look rather than trusting memory, coverage changes fast).
- **Bhashini** (Government of India's national language mission) — API access to Indic ASR/MT/TTS, actively expanding NER language support.

Since no public dataset covers NER languages or NER-specific reminiscence content, your data strategy has three tracks:

Track A — Bootstrap without data (Phase 1, ship first)

- Rule-based adaptive difficulty: simple staircase algorithm (raise difficulty after N correct, lower after N wrong) — no ML needed, works from day one.
- Reminiscence content: crowd-source/manually curate — festival images (Bihu, Hornbill, Chapchar Kut), regional folk songs, local landmarks, common foods. This is a content-sourcing task, not a data-science task: partner with local cultural bodies, ARDSI Guwahati, community elders' groups.
- Games: build 8–12 templates, populate with localized content packs per state/language.

Track B — Your own telemetry becomes your dataset (Phase 2, months 3–12)

- Every game session logs: reaction time, accuracy, error type, task difficulty, time-of-day, session length.
- This *is* your labeled dataset for the decline-detection model, once you have enough users and enough time elapsed to see trends (this needs longitudinal data — months, not days).
- Pair with periodic caregiver-reported MMSE/MoCA scores (done by a partnering clinician) as ground-truth labels — this is how you turn your app usage logs into a supervised dataset.

Track C — Targeted small-scale collection for NER speech (parallel effort)

- Partner with a college/hospital (e.g., Guwahati Medical College, which already participates in LASI-DAD) to record a small consented corpus: elderly speakers doing simple picture-description or naming tasks in Assamese/Manipuri/Bodo — even 50–100 speakers gives you enough for a feasibility pilot and fine-tuning a multilingual ASR model.

- IRB/ethics approval and informed consent (via caregivers, since your subjects are cognitively vulnerable) is mandatory here — this isn't optional paperwork, it's the same standard used in the SMRUTHI and LASI-DAD studies you're building on.

The core idea is to build a **localized, offline-first cognitive support platform for elderly users in North-East India**, especially people with mild cognitive impairment or early-stage dementia support needs.

The platform is **not a dementia cure or diagnostic tool**. Its immediate purpose is to help with cognitive engagement, daily memory assistance, culturally familiar reminiscence, and caregiver monitoring. The more advanced AI components come later, once enough real usage data exists. That framing matches the evidence and caution in the source document.

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1. Who uses the system?

There are mainly two users.

Elderly user

Uses the mobile app for:

- cognitive games
- reminders
- family memories
- culturally familiar reminiscence content
- eventually voice interaction

Caregiver

Uses a separate dashboard to:

- monitor game participation
- see performance trends
- create reminders
- upload family photos/content
- see alerts or meaningful long-term changes

A clinician can be added later, but we are **not making the clinician portal a core MVP requirement**.

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2. Main features we are building

A. Cognitive games

The elderly user gets short games that exercise different cognitive abilities.

Examples:

Game type	What it exercises
Matching pairs	Memory
Tap/find target	Attention + reaction speed
Arrange steps	Executive function
Picture naming	Language/recall
Simple arithmetic	Calculation
Path/maze tasks	Visuospatial ability

The source proposes cognitive games across memory, attention, language, executive function, visuospatial ability and arithmetic.

For example:

1. Memory matching
2. Attention/reaction game
3. Sequencing game
4. Picture naming

3. Every game collects performance data

Every session should record things like:

user_id
game_type
difficulty
accuracy
reaction_time
errors
hints_used
session_duration
completed_or_quit
timestamp

This is extremely important because the app is not only providing cognitive activities.

It is also gradually building a **longitudinal behavioral dataset**.

For example:

Week 1
Accuracy = 84%
Reaction = 2.8s

Week 4
Accuracy = 81%
Reaction = 3.0s

Week 10
Accuracy = 72%
Reaction = 4.1s

That allows us to study changes over time.

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4. Adaptive difficulty

For version 1, we **do not use reinforcement learning yet**.
We use a straightforward rule-based system.

Example:

3 strong rounds
↓
increase difficulty

2 poor rounds
↓
decrease difficulty

otherwise
↓
keep same difficulty

This makes the game personalized from day one without needing training data.

Later, once we accumulate real user telemetry, we can replace or augment this with reinforcement learning/contextual bandits.

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5. Future reinforcement-learning system

Later, the RL system learns what difficulty or activity is appropriate for each user. (Track B gameplay data)

Conceptually:

STATE
recent accuracy
reaction time
current difficulty
recent errors
session behaviour

↓

ACTION
choose next difficulty/game

↓

USER PLAYS

↓

REWARD
did the user perform well
without the task being too easy?

So the goal isn't:make the user score 100%.

Because then the system would just give easy games.

The goal is:keep the person appropriately challenged and engaged.

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6. Reminiscence feature

This is one of the strongest parts of the project.

Instead of generic images, the app contains **culturally and personally familiar content**.

For example:

Bihu
Hornbill Festival
regional foods
traditional clothing
local landmarks
regional songs
family photographs
childhood places
old wedding pictures

The source specifically highlights culturally localized reminiscence as an important opportunity for the NER context.

The caregiver can also upload personal material.

For example:

Photo: Family wedding
Year: 1998
People: Daughter, son, husband
Location: Guwahati

The app could show:

> "Do you remember who this is?"

or:

> "Where was this photograph taken?"

This creates a nice overlap between:

Cognitive gaming+Personal memories+Reminiscence

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7. Memory assistance

The app also assists the user with everyday life.

Core features:

Reminders

Medicine at 8 PM
Lunch at 1 PM
Doctor appointment tomorrow

Reality orientation

Display:

Wednesday
26 August 2026

Good evening.

You are at home.

Next reminder:
Medicine at 8 PM

Family recall

The caregiver can upload:

Photo
Name
Relationship
Short description

Then the elderly user can revisit familiar people.

The source proposes these daily-living support features directly.

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8. Caregiver dashboard

The caregiver gets a normal web dashboard.

They might see:

Sessions this week: 5
Average memory accuracy: 78%
Reaction time: 3.2s
Games completed: 12
Missed reminders: 1

And graphs such as:

Memory Accuracy
90%
80%
70%

They can also:

- create reminders
- upload family photographs
- manage reminiscence content
- see activity history

Later, meaningful performance deviations could generate alerts.

The source proposes progress trends, alerts, medication logs and caregiver support.

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9. Cognitive trend monitoring

This is different from reinforcement learning.

RL answers:

> “What game/difficulty should I give this user next?”

Trend monitoring asks:

> “Is this person's performance changing over time?”

Initially we can calculate simple statistics:

7-day accuracy average
30-day accuracy average

average reaction time
reaction-time change

error-rate change
game completion rate

Then compare:

CURRENT PERFORMANCE
vs
PERSONAL BASELINE

For example:

> Memory-game response time has increased 18% compared with the user's normal baseline.

We **do not say**:

> Dementia probability: 83%.

Not without proper clinical validation.

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10. Long-term decline-detection ML

Eventually, this could become a genuine research component.

You would have:

Gameplay telemetry
+
longitudinal data
+
periodic validated clinical assessments

Then you could study whether:

reaction time ↑
accuracy ↓
errors ↑
task abandonment ↑

correlate with clinically assessed cognitive changes.

That is where supervised ML models such as:

XGBoost
Random Forest
time-series models

could potentially come in.

But that is **not v1**.

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11. Voice assistant

Eventually we want the app to be voice-first because typing and reading may be difficult for some elderly users.

Architecture:

User speaks
↓
ASR
Speech → Text
↓
Assistant / backend
↓
Response text
↓
TTS
Text → Speech
↓
User hears answer

For example:

> “When is my medicine?”

The system could reply:

> “Your evening medicine is scheduled for 8 PM.”

NER language support is one of the hardest but most valuable parts of the proposed platform.

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12. Language scope

We should **not attempt all NER languages initially**.

For MVP:

> **one regional language first — Assamese is a sensible candidate.**

Potential later expansion:

Assamese
↓
Manipuri/Meitei
↓
Bodo
↓
other regional languages

The original document also recommends starting with only a few languages because localization at scale is a major difficulty.

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13. Speech-data research track

If we eventually want genuinely good NER speech support, we may need our own small regional speech corpus.

For example:

50–100 elderly Assamese speakers

Tasks:
picture description
object naming
simple conversation
short memory tasks

We would collect:

audio

+
human verified transcription
+
language
+
task metadata

Then use that to evaluate or fine-tune an **existing pretrained multilingual model**.

Not train a speech model from scratch.

This part requires proper ethics/consent if collected from real vulnerable participants.

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14. Offline-first design

This is an important architectural requirement because connectivity may be unreliable in rural and hill regions. The source explicitly identifies connectivity as a challenge.

Normally:

App
↓
Internet
↓
Backend

means:

No internet = broken app

We want:

```
     INTERNET?
      /     \
    YES      NO
     ↓        ↓
  FastAPI   SQLite
     ↑        |
     └── Sync ──┘
```

So even offline, the user can:

- play games
- see cached memories
- receive locally scheduled reminders
- generate gameplay results

When internet returns:

Phone
↓
unsynced results
↓
FastAPI
↓
PostgreSQL

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15. Finalized mobile technology

React Native

For the mobile application.

Why:

- Android/iOS cross-platform
- good ecosystem
- suitable for an MVP
- handles mobile UI, notifications, storage, etc.

The UI needs to be:

- large buttons
- simple navigation
- high contrast
- minimal text
- eventually voice-driven

The source specifically recommends elderly-friendly simple UI and voice-first navigation.

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16. Local mobile database

SQLite

SQLite sits directly on the user's phone.

It stores:

cached games
reminders
basic user profile
recent results
cached reminiscence information
pending sync events

This is what enables offline-first functionality.

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17. Backend

FastAPI + Python

FastAPI handles all server communication.

Architecture:

```
React Native
    ↓
FastAPI
  ↙   ↘
Database   AI/analytics
```

It can expose endpoints such as:

POST /game-sessions

GET /users/{id}/analytics

POST /reminders

GET /reminders

POST /memories

GET /caregiver/{id}/patients

We prefer FastAPI largely because the later analytics/ML stack will also be Python.

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18. Main database

PostgreSQL

This stores structured server-side data.

Likely tables:

users

caregivers

caregiver_user_links

games

game_sessions

reminders

memory_items

performance_metrics

alerts

sync_events

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19. Media storage

Family photos, music and videos shouldn't be stored directly inside PostgreSQL.

Use object storage such as:

AWS S3
or
Cloudflare R2
or
Supabase Storage

Then:

PostgreSQL
stores:
photo name
user
description
URL

Object storage
stores:
actual .jpg/.mp3/.mp4

The original architecture proposal also separates cloud media storage from the normal database.

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20. Caregiver frontend

React / Next.js

This will be the caregiver web dashboard.

Architecture:

React Native elderly app
↓
FastAPI
↑
React/Next caregiver dashboard

Both interfaces communicate with the same backend.

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21. Overall architecture

```
                ELDERLY USER
                      |
                      ↓
               React Native App
                      |
      ┌───────────────┼───────────────┐
      ↓           ↓          ↓
   Games      Reminders   Memories
      |
      ↓
Rule-based adaptation
      |
      ↓
   SQLite
 Local/offline data
      |
      | sync
      ↓
   FastAPI
      |
  ┌───┴────────────────┐
  ↓             ↓
PostgreSQL    Object Storage
  |             |
gameplay      photos/audio
telemetry       video
  |
  ↓
Analytics Engine
  |
 ┌┴────────────────┐
 ↓          ↓
Trends     Future RL/ML
  |
  ↓
Caregiver Dashboard
React / Next.js
```